

The background of the slide is split diagonally. The top-left portion shows a photograph of a landscape with solar panels in the foreground, a line of evergreen trees, and a grassy hill under a blue sky with white clouds. The bottom-right portion is a solid light gray.

Lecture 5: Inverter Control

ELEN0445 - Microgrids

Bastien Ewbank Bertrand Cornélusse

12 March 2026

Motivation of Today

Room 0/64, Building B37, Educational Microgrid Setup



Motivation of Today

Single-Wire Equivalent Diagram: How do these devices work?



Outline

Voltage Source Converter (VSC)

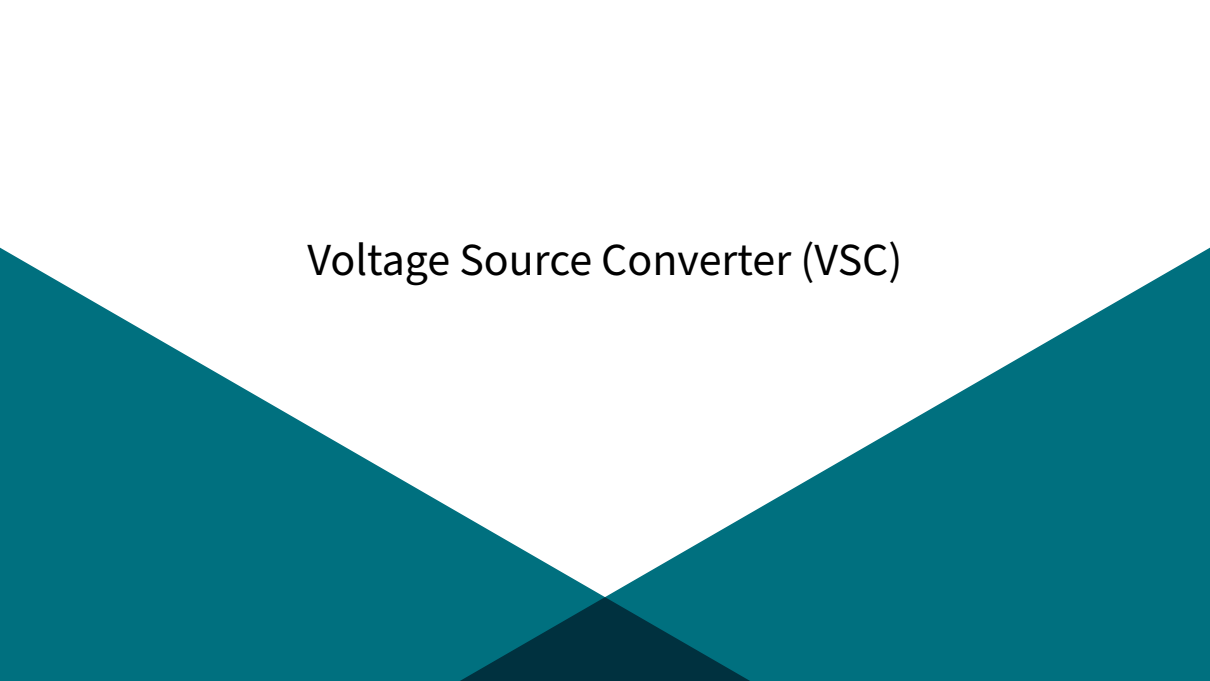
Grid Forming or Grid Following?

Control Design

Softstart and Islanding

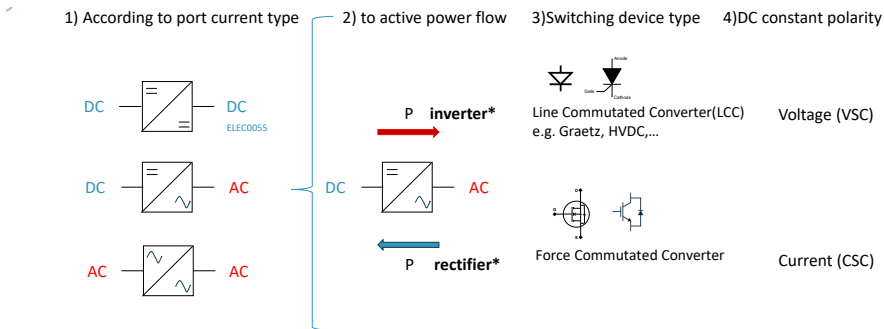
Summary

Voltage Source Converter (VSC)

The background of the slide features a white upper section and a teal lower section. The teal section is composed of two large triangles that meet at a point at the bottom center, creating a V-shape. The text 'Voltage Source Converter (VSC)' is centered in the white area above this teal section.

Classification of Converters

According to Mohan et al. and Yazdani & Iravani [1, 2]



According to Ned Mohan 's and Amirnaser Yazdani's books [1-2]

*In some references, 'inverter', resp. 'rectifier', are sometimes rather linked to the step up, resp. step down, behavior of the Voltage between AC and DC

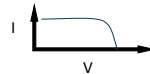
How Would You Classify Our Devices?



- 3 ports : 2 AC – 1DC $\rightarrow$ AD/DC
- Power can flow in both direction with respect to DC port $\rightarrow$ Bidirectional converter or Inverter and rectifier (several names valid)
- Probably* at least few IGBT or MOSFET in the converter circuit $\rightarrow$ Force Commutated Converter
- A battery can be model as a constant voltage source, and as P flows in both direction current will have + and - polarity $\rightarrow$ VSC



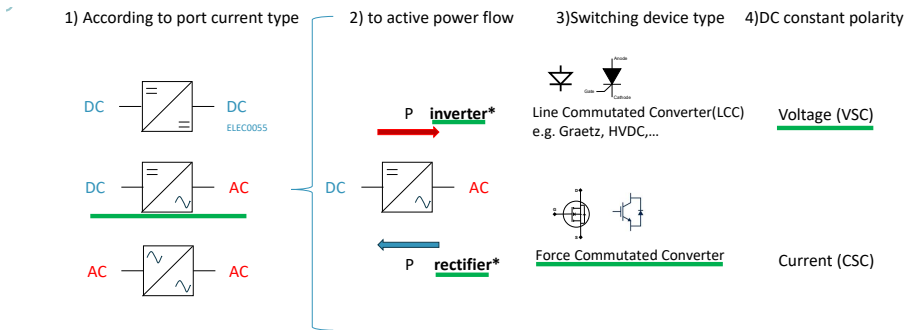
- 2 ports : 1 AC – 1DC $\rightarrow$ AD/DC
- Power can flow in both direction with respect to DC port **but lighting the sky with bad LED is a very useless usage ;)**, so... $\rightarrow$ Inverter
- Probably* IGBT or MOSFET based inverter circuit $\rightarrow$ Force Commutated Converter
- PV panels ?? $\rightarrow$ Not so obvious for now ...



*Manufacturers do not publish the schematic of their converter topology, and reverse

Classification of Converters: Focus on VSC

Forced commutated voltage source converters (Scope of today)



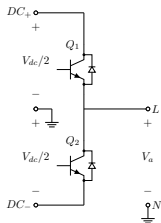
According to Ned Mohan's and Amirnaser Yazdani's books [1-2]

Scope of today

*In some references, **inverter**, resp. **rectifier**, are sometimes rather linked to the step up, resp. step down, behavior of the Voltage between AC and DC

AC-DC Topologies in Practice

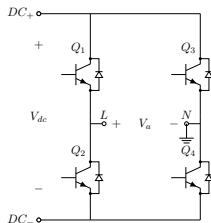
Half-Bridge (1ϕ)



2 switches

Requires split DC bus

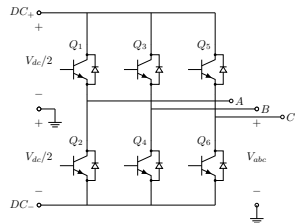
Full-Bridge (1ϕ)



4 switches (H-bridge)

Bipolar / unipolar PWM

Three-Phase Bridge (3ϕ)



6 switches

Standard 3ϕ VSC

Modern Multilevel Topologies

NPC (Neutral Point Clamped), T-Type, MMC (Modular Multilevel Converter), etc. (beyond the scope of this lecture).

AC-DC VSC for Grid Applications

Motivation

- ▶ **AC:** Historically, the electricity grid was built based on AC systems.
- ▶ **DC:** PV panels are DC sources that are available worldwide and represent affordable renewable energy. Batteries are also DC, and many other sources traditionally based on AC are interfaced through back-to-back AC-DC-AC converters.
- ▶ **VSC:** Controlling power flow is easier by modulating voltage than with a current source, as the grid connection point behaves essentially as a voltage source.

What About Control?

All preceding classification criteria were based solely on the **power electronics topology** (hardware components through which power flows).

→ **How do we control these switches to manage voltage, current, and power?**

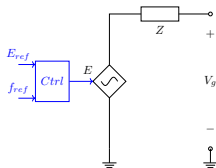
Grid Forming or Grid Following?

Grid Forming (GFOR) vs. Grid Following (GFOL)

- ▶ GFOR and GFOL are **control strategies**, not different converter topologies!
- ▶ The same physical converter can switch between GFOR and GFOL when appropriate.

Grid Forming (GFOR)

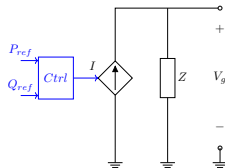
« I am forming a Voltage source »



- ▶ Acts as an **ideal AC Voltage Source** (E_{ref}, f_{ref})
- ▶ Similar in role to a synchronous generator
- ▶ Requires a large, controllable DC source (battery, fuel cell)

Grid Following (GFOL)

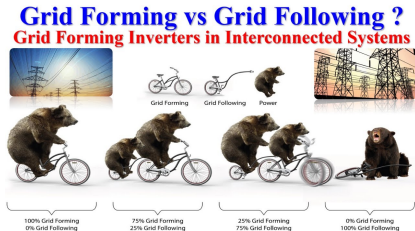
« I am following a Voltage source »



- ▶ Acts as an **AC Current Source** (P_{ref}, Q_{ref})
- ▶ Must sense a low-distortion grid voltage (V_g, f_g) to synchronize
- ▶ Standard for renewable sources (e.g. PV with MPPT)

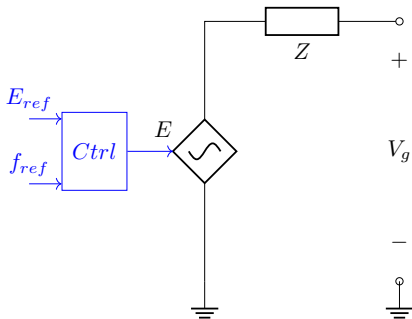
Comments on GFOR vs. GFOL

- ▶ Nowadays, almost every distributed generator (DG) like PV or wind turbines is **GFOL controlled**.
- ▶ As conventional synchronous machines retire ($\searrow$) and converter-interfaced DG grows ($\nearrow$), the power grid becomes **weaker** (lower physical inertia, lower short-circuit ratio).
- ▶ Extensive research is focused on **GFOR deployment**:
 - ▶ Power semiconductor current limits require fast current-limiting algorithms.
 - ▶ Basic GFOR lacks physical rotor inertia; synthetic inertia / droop control must be emulated.
- ▶ Advanced concepts: **Grid Supporting Converters, Virtual Synchronous Machines (VSM)**, etc. [3, 4].

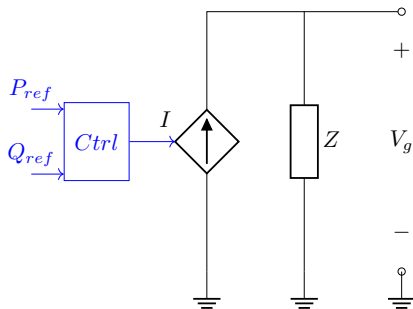


Control Architecture

GFOR Control Strategy



GFOL Control Strategy



What is inside these « Ctrl » blocks?

→ Let's discover the theory and observe them with a Typhoon HIL live demo!

Control Design

The background of the slide features a white central area where the text is located. This central area is framed by two large teal-colored triangles that point towards each other from the left and right sides, meeting at a point at the bottom center. The overall effect is a stylized, modern graphic design.

Clarke & Park Transformation

Clarke Transformation ($abc \rightarrow \alpha\beta 0$):

$$T_c = \frac{2}{3} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

Rotation Matrix of angle ωt :

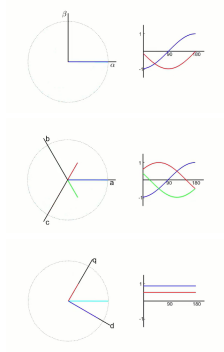
$$R(\omega t) = \begin{bmatrix} \cos(\omega t) & \sin(\omega t) & 0 \\ -\sin(\omega t) & \cos(\omega t) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Park Transformation ($T_p = R(\omega t)T_c$):

$$T_p(\omega t) = \frac{2}{3} \begin{bmatrix} \cos(\omega t) & \cos(\omega t - \frac{2\pi}{3}) & \cos(\omega t + \frac{2\pi}{3}) \\ -\sin(\omega t) & -\sin(\omega t - \frac{2\pi}{3}) & -\sin(\omega t + \frac{2\pi}{3}) \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

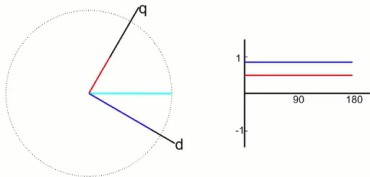
Signals across coordinate frames:

$$V_{abc}(t) \xrightarrow{\text{Clarke}} V_{\alpha\beta 0}(t) \xrightarrow{\text{Rotation}} V_{dq0}(t)$$



Usefulness for AC Systems

$$V_{abc}(t) = \begin{bmatrix} V_a(t) \\ V_b(t) \\ V_c(t) \end{bmatrix} \xrightarrow{T_p(\omega t)} V_{dq0}(t) = \begin{bmatrix} V_d(t) \\ V_q(t) \\ V_0(t) \end{bmatrix}$$



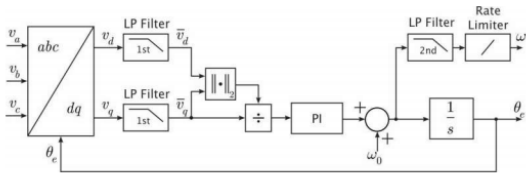
- ▶ The ***dq0* reference frame** is analogous, in terms of purpose, to the **phasor** ($\underline{V}$) concept used for sinusoidal steady-state AC circuits.
- ▶ It converts balanced three-phase time-varying sinusoidal signals into **constant DC quantities** in steady state.
- ▶ **Control theory advantage:** Time-invariant signals allow the direct application of standard linear controllers (e.g. **PI / PID controllers**) with zero steady-state error.

Phase-Locked Loop (PLL)

- ▶ To compute the Park transform $T_p(\omega t)$, the angle $\theta(t) = \omega t$ must be known.
- ▶ The **Phase-Locked Loop (PLL)** tracks the grid frequency and phase angle (essential for grid-following converters).

Three-phase PLL operating principle:

1. Apply Park transform $T_p(\hat{\theta})$ to measured grid voltages V_{abc} .
2. Regulate $V_q \rightarrow 0$ using a PI controller.
3. When $V_q = 0$, the rotating d -axis is aligned with the grid voltage vector.
4. Integrator outputs tracked angle $\hat{\theta} = \omega t$.



Current Controller: Governing Equations

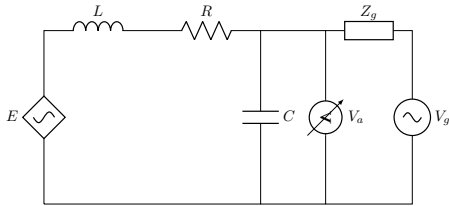
- ▶ Actuators of a VSC are switching devices producing a voltage waveform $E(t)$.
- ▶ How do we control the current if we only command voltage?
- ▶ → Using **Ohm's and Faraday's laws** across the grid filter (L, R):

$$E(t) - V_a(t) = L \frac{di(t)}{dt} + Ri(t)$$

- ▶ Moving to the rotating $dq0$ reference frame ($V_{abc} = T_p^{-1} V_{dq0}$):

$$T_p^{-1}(\omega t) = \begin{bmatrix} \cos(\omega t) & -\sin(\omega t) & 1 \\ \cos(\omega t - \frac{2\pi}{3}) & -\sin(\omega t - \frac{2\pi}{3}) & 1 \\ \cos(\omega t + \frac{2\pi}{3}) & -\sin(\omega t + \frac{2\pi}{3}) & 1 \end{bmatrix}$$

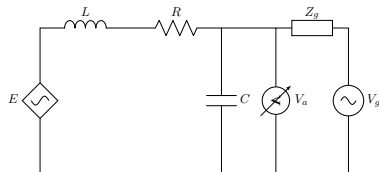
Single-Phase Equivalent AC Circuit:



Current Controller: Decoupled dq Control

Differentiating in the rotating frame yields cross-coupling terms between d and q axes:

$$E_d(t) = \underbrace{L \frac{di_d(t)}{dt} + Ri_d(t)}_{\text{PI output } u_d} - \underbrace{L\omega i_q(t)}_{\text{cross-coupling}} + \underbrace{V_d(t)}_{\text{feedforward}}$$
$$E_q(t) = \underbrace{L \frac{di_q(t)}{dt} + Ri_q(t)}_{\text{PI output } u_q} + \underbrace{L\omega i_d(t)}_{\text{cross-coupling}} + \underbrace{V_q(t)}_{\text{feedforward}}$$



- **PI Controller terms:** Regulate tracking errors $(I_d^* - I_d)$ and $(I_q^* - I_q)$.
- **Decoupling & Feedforward:** Cancel cross-coupling $(\pm L\omega I)$ and grid voltage disturbance (V_{dq}) .

Softstart and Islanding

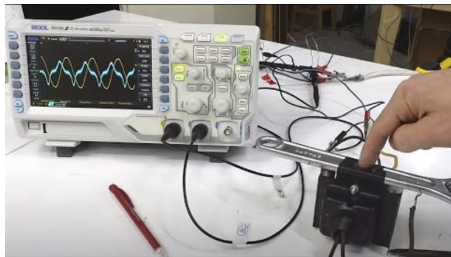
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Softstart and Islanding: Managing Transients

Key Takeaway

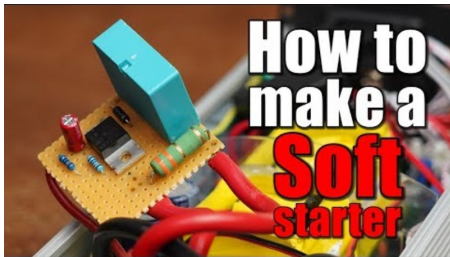
When designing or commissioning converters, **never overlook transient electrical behaviors** before closing relays or connecting circuits!

Magnetic Inrush Transient



Transformer core saturation upon energization

Capacitive Inrush Transient



Uncontrolled charging of DC-link capacitors

Summary

The background of the slide features a minimalist design with teal-colored geometric shapes. Two large triangles, one on the left and one on the right, point towards the center. These triangles overlap at the bottom, where they meet a darker teal triangular shape that points upwards.

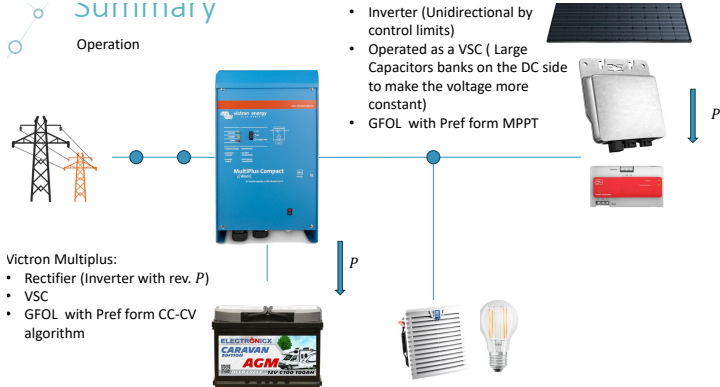
Key Takeaways

- ▶ **VSCs** are AC/DC converters with constant DC voltage polarity, modulating the desired waveform on the AC side.
- ▶ To operate properly as an inverter, a VSC requires $V_{dc} \geq \max_t V_a(t)$ (for full bridge).
- ▶ Active power P can flow from AC to DC uncontrolled via the anti-parallel body diodes if $V_{dc} \leq \max_t V_a(t)$.
- ▶ **GFOR** acts as a **Voltage Source**; **GFOL** acts as a **Current Source**.
- ▶ In practice, simple open-loop GFOR exists only in islanded microgrids with ideal sources. Real grid-connected GFOR converters implement P and Q current limitations to protect semiconductors.
- ▶ A **PLL** is a dedicated closed-loop control unit that tracks the frequency and phase angle of an AC voltage signal.

Summary: Grid-Connected Operation

Room 0/64 B37 Educational Microgrid

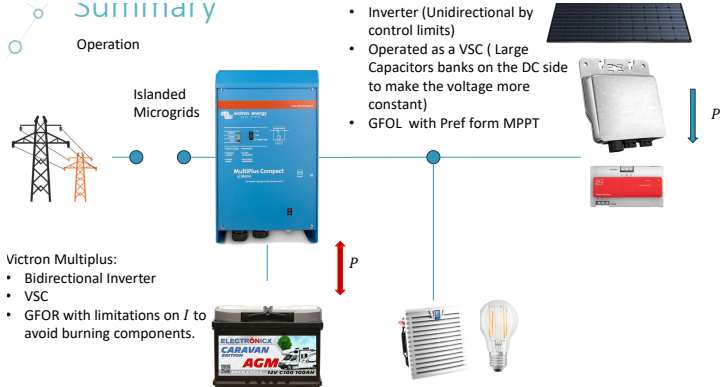
Summary Operation



Summary: Islanded Microgrid Operation

Room 0/64 B37 Educational Microgrid

Summary Operation



References

1. Mohan, N., Undeland, T. M., & Robbins, W. P. (1995). *Power Electronics: Converters, Applications, and Design* (2nd ed.). John Wiley & Sons.
2. Yazdani, A., & Iravani, R. (2010). *Voltage-Sourced Converters in Power Systems: Modeling, Control, and Applications* (1st ed.). Wiley. <https://doi.org/10.1002/9780470551578>
3. Rocabert, J., Luna, A., Blaabjerg, F., & Rodríguez, P. (2012). Control of Power Converters in AC Microgrids. *IEEE Transactions on Power Electronics*, 27(11), 4731–4746. <https://doi.org/10.1109/TPEL.2012.2199334>
4. Zhang, H., Xiang, W., Lin, W., & Wen, J. (2021). Grid Forming Converters in Renewable Energy Sources Dominated Power Grid: Control Strategy, Stability, Application, and Challenges. *Journal of Modern Power Systems and Clean Energy*, 9(6), 1239–1256. <https://doi.org/10.35833/MPCE.2021.000257>